

SmartHire GenAI -- Capstone Final Project Report

Project Title: SmartHire GenAI -- Resume Matching & AI Career Mentor

Technologies: Google Gemini 3.8/3.6 Flash, Text Embeddings, FAISS Vector Database, LangChain RAG, Pydantic, Streamlit

Date: September 2026

1. Executive Summary

SmartHire GenAI is an intelligent career portal that bridges the gap between candidate resumes and modern employment opportunities. Unlike traditional keyword-matching Applicant Tracking Systems (ATS), SmartHire uses generative AI, vector embeddings, and Retrieval-Augmented Generation (RAG) to provide:

1. Automated Structured Resume Extraction into typed, validated schemas with Human-in-the-Loop review.
2. Semantic Job Matching using cosine similarity over a pre-indexed vector corpus in FAISS.
3. Targeted CV Improvement Suggestions targeting missing skills, weak bullet points, and rewritten summaries.
4. AI Career Mentor Chatbot grounded strictly in verified career notes, role roadmaps, and interview standards with zero tolerance for hallucinations.
5. Kaggle 24-Industry Resume Explorer & Dynamic JSON Generator capable of parsing thousands of diverse resumes on demand.

2. Design Choices & Architecture

A. LLM & Embedding Selection

- Primary LLM: gemini-3.8-flash (with automated fallback to gemini-3.6-flash for high throughput and rate-limit resilience).
- Embedding Engine: gemini-embedding-001 (3072-dimensional normalized vectors) coupled with an in-memory normalized fallback hash vectorizer ensuring offline/test reliability.
- Structured Outputs: Leveraging Gemini's native response_json_schema backed by Pydantic models (ResumeProfile, DynamicResumeJSON) to enforce clean, predictable data contracts.

B. Vector Storage & Retrieval

- Engine: Facebook AI Similarity Search (faiss-cpu) using IndexFlatIP on L2-normalized vectors (mathematically equivalent to cosine similarity).
- Dual Vector Spaces:
 1. Job Index (vectorstore/jobs/): Indexes titles, skills, and full descriptions of industry roles.
 2. Mentor Index (vectorstore/mentor/): Indexes chunked career guides, roadmaps, and interview prep notes.

C. Human-in-the-Loop (HITL) State Machine

- Strict architectural guardrail: candidates must review and approve their extracted profile before downstream semantic matching or CV suggestions execute.
- Prevents cascading hallucinations from unverified LLM parser extractions.

D. Grounded RAG & Responsible AI Guardrails

- Lexical & Semantic Grounding: Mentor answers are conditioned exclusively on retrieved chunks.
- Strict Out-of-Scope Refusal: When asked questions outside career guidance (or unsupported facts), the mentor enforces: "I don't know based on the available documents."
- Safety Guardrails: Pre-execution regex and semantic filters intercept prompt injections, secret exfiltration, and fraudulent resume tailoring attempts.

3. What Worked Well

1. Semantic Search Superiority: Candidate profiles match roles based on conceptual skill overlap (e.g. PyTorch matching Deep Learning positions) rather than literal token matches.
2. Deterministic Fallbacks: The multi-tiered architecture (Gemini 3.8 -> Gemini 3.6 -> Heuristic Parser) achieved a 100% test suite pass rate (30/30 unit & integration tests) even under API quotas.
3. Calm SaaS User Experience: Converting fragmented tab interfaces into a sequential 5-step workspace with native Light/Dark themes created a serious, presentation-ready product.
4. Dynamic JSON Generation: Processing resumes dynamically from Kaggle's 2,484-resume dataset unlocks instant multi-industry demonstrations without storage overhead.

4. Limitations & Future Roadmap

1. Static Job Corpus: The current job catalog is pre-collected from Kaggle rather than live web scraping (complying with platform Terms of Service). Future iterations could integrate official LinkedIn or Indeed APIs.
2. Multi-Page Resume Complexity: Highly non-standard multi-column resume PDFs occasionally require human adjustments during the HITL review step.
3. Cross-Encoder Re-ranking: While bi-encoder FAISS retrieval achieves high recall, adding a cross-encoder re-ranking stage (e.g., FlashRank or BGE-Reranker) could further refine top-3 ranking precision.